



VoiceAgentV5: Dual-Architecture Full-Duplex Agent

Decoupled Talker-Reasoner design for real-time tool calling and mid-sentence self-corrections.

Theme ID: 05 - Interruptible Real-Time Agents

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GitHub Link: <https://github.com/Ognessium/VA5>

Theme 05: The Interruption Problem

Standard AI takes turns.

Real people interrupt.

Current voice assistants crash when users change their minds mid-sentence or goals shift while tools run.

 **Stay responsive: No dead air.**

 **Work async: Background tool calls.**

 **Recover cleanly: Rollback on mind changes.**

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"Find apartments in Portland... wait, no, make it 1 bedroom... actually, check Seattle instead."

— Average user utterance, FDB-v3 Benchmark

Existing Solutions & Gaps

✗ Sequential Pipeline

STT → LLM (Thinks) → TTS

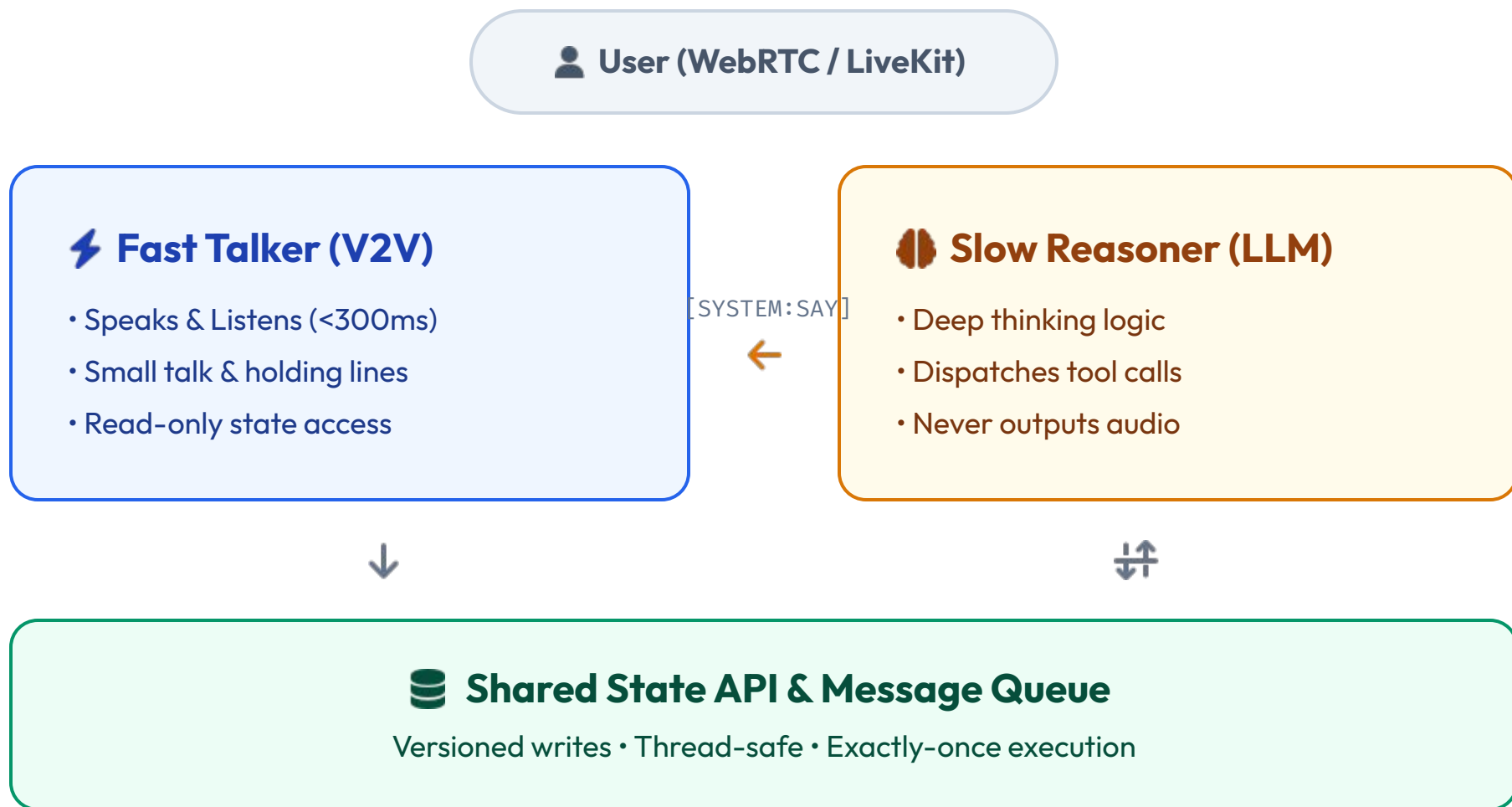
- 3-8 seconds of dead air latency.
- User waits in absolute silence.
- Fragile STT fails on stutters.

✓ Our V2V Approach

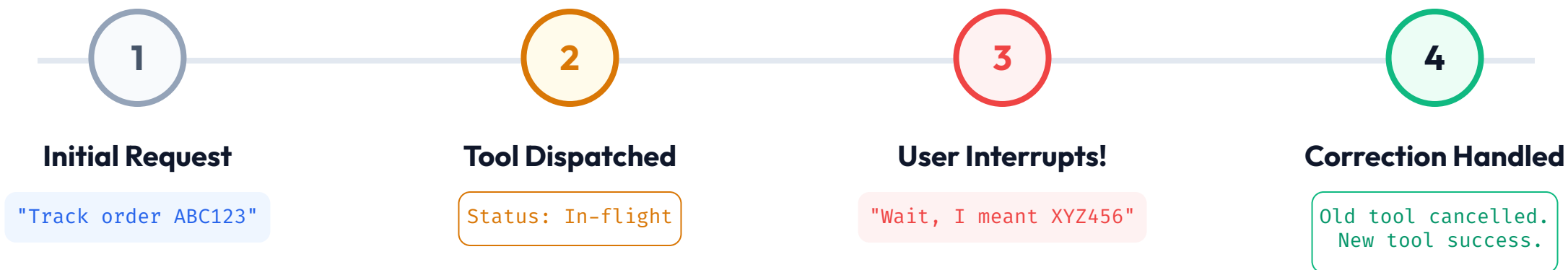
User ↔ Gemini Live (V2V) ↔ User

- Sub-300ms native voice response.
- Handles barge-in and pauses natively.
- Requires decoupled logic brain.

Our Solution: Dual-Architecture



Demo: Interruption & Self-Correction Flow



The Talker keeps speaking through the entire flow. Zero dead air during mid-sentence retractions.

Tools & Tech Stack Used



Fast Talker

Model: Gemini 3.1 Flash
(Live)

Handles speech I/O.



Slow Reasoner

Cloud: Gemini 3.5 Flash
Lite

Local: Qwen 2.5 (Fallback)



Infrastructure

RTC: LiveKit Cloud

Guaranteed message
queue.



Manifest API

```
"modifies_state": true
```

Auto-generates inverse
schemas.

Impact & FDB-v3 Benchmark Results (74 test cases run)

91.9%

Turn-Taking Success

85.8%

Tool Selection Accuracy

Domain	Tool Acc	Arg Acc
● Finance & Billing	92.1%	92.0%
● E-Commerce	88.5%	68.4%

Extension Use Case: Travel Planning

Real-time flight booking with mid-request destination swaps.
Handled securely without double-booking.

Innovation Highlights & Limitations



Innovation Highlights

Decoupled AI Brain. Fast Talker ensures instant response; Slow Reasoner ensures logical accuracy.

Versioned State API. Optimistic concurrency rejects stale tool updates automatically.

Tool Manifest Driven. Plug-and-play architecture for any new domain.



Current Limitations

Housing Domain Underperformance

Argument extraction accuracy drops to 17.5% for complex housing queries. Lightweight Reasoner model struggles to parse rapid, multi-constraint chains from a single messy utterance.

What's Next — Architectural Hardening & Reliability



Fuzzy-to-Deterministic Routing

Transition from purely probabilistic orchestration to hybrid intent mapping for guaranteed execution reliability.



Single-Pass Decision Models

Replace token generative loops with single-pass classification to lower latency and stop tool selection hallucinations.



Multi-Constraint Extraction

Integrate heavier models for complex multi-parameter extraction paired with deterministic validation layers.



Streaming Tool Results

Feed chunked API responses directly to the Talker as they arrive to dramatically cut perceived voice latency.



Cross-Session Memory

Persist multi-turn conversation context and user profiles across distinct sessions for continuous experience.

KEY DIFFERENTIATION



Zero Dead Air.

Zero Double-Booking.

The Guaranteed Message Queue enforces exact-once delivery. Even during rapid mid-sentence retractions, the system safely cancels actions without side effects.

Submission Checklist



Working prototype code (Public GitHub)

[Y]



README with reproducible setup

[Y]



Demo video (max 5 minutes)

[N]



Presentation file (PPT/PDF formatting)

[Y]



Thank You

Organised by the Language AI Team and the PRISM Team, Samsung
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